SiteSaathi (□□□□ □□□□)

Project Documentation: System Overview, Stakeholder Roles & Application Flow

Target Audience: Civil, Plumbing & Electrical Contractors (Thekedars)	Interface Languages: Bilingual (Hindi / English)
Operational Mode: 100% Offline-Capable Local-First	Core Differentiators: Voice-to-Form & Photo Register OCR

1. Executive Summary

SiteSaathi is a specialized digital site assistant designed for contractors and site supervisors. In the everyday Indian contracting industry, vital project records are traditionally jotted on loose notebook pages, whatsapp messages, or kept in memory. This leads to common financial leakages: forgotten tool rentals accumulating extra fees, delayed material credit (Udhaar) settlements, dispute-prone worker wage balances, and lost profit margins.

SiteSaathi digitizes this entire lifecycle with extreme simplicity. It provides big touch targets, high contrast, bilingual Hindi/English support, and removes manual typing friction by allowing contractors to speak site details or take a photo of their handwritten paper register.

2. Stakeholder Roles & User Personas

A. The Contractor / Site Manager (Primary User)

The primary app operator. The application adjusts its entire theme, metrics, units, and fast-templates according to their specific trade:

- General Construction Contractor: Manages civil structures, masonry, plaster, shuttering plates (4x4), concrete mixers, and JCB excavators. (Safety Orange Theme)
- Plumbing & Sanitary Contractor: Manages water lines, drainage pipes (CPVC/UPVC), fittings, threaders, core cutters, and sanitary fixtures. (Cool Aqua Theme)
- Electrical & Wiring Contractor: Manages concealed conduits, wire bundles, distribution boxes, cable pullers, multimeters, and switches. (Electric Amber Theme)

B. Daily Labour & Site Workers (Managed Staff)

- Masons / Mistry (□□□□□□ / □□□□□□□□): Skilled technical workforce.
- Helpers / Beldar (□□□□□ / □□□□□□): General site support labour.
- Specialized Trades: Plumbers, Electricians, Plasterers, Welders, Carpenters, Painters, and Site Supervisors.
- Attributes Tracked: Full day / Half day / Absent / Overtime status, daily wage rate, on-site cash advances, and unpaid wage balances.

C. Vendors & Suppliers (□□□□□□□□ / □□□□□□□)

- Material stores, tool rental agencies, and machinery contractors.
- Tracked for purchase invoices, payment statuses, and outstanding credit balances (Udhaar / □□□□).

D. Clients / Project Owners

- Property owners or developers who fund the projects. Tracked for contract budget, milestone progress, and financial statement exports.

3. End-to-End Application Workflow

Step 1: Onboarding & Trade Pick	Contractor inputs their name and phone, then chooses their operating language (Hindi or English) and trade type (Construction, Plumbing, or Electrical).
Step 2: Site Creation (Manual or Voice)	Contractor creates a project. They can fill fields manually or tap the microphone to speak in Hindi/English (e.g. '□□□□ □□ □□□ □□□□□ □□□□, □□□ 8 □□□'). The speech parser auto-fills the record.
Step 3: Daily Field Record Logging	On active work days, contractor logs updates across 4 core operational pillars: 1. Labour Register: Mark attendance & wages, or photograph paper muster notebook for auto-digitization. 2. Tools Tracker: Track tool dispatch and return deadlines; flags overdue items. 3. Materials Log: Log purchases with quantities, unit prices, and mark paid vs. Udhaar. 4. Machinery & Fuel: Log JCB/Mixer operating hours and diesel fuel consumption.
Step 4: Pending Dues & Financial Ledger	Decouples physical job progress from payment status. When a site's work finishes, unpaid vendor bills and tool dues stay highlighted in a dedicated tracker until settled.
Step 5: Automated Smart Insights	Rule engine continuously flags risks: overdue tools building up rental costs, high-outflow sites exceeding budget, and pending dues on completed sites.
Step 6: Reports & PDF Export	Generates client financial statements with cost breakdowns and CSV attendance muster rolls for accountants.

4. Core Real-World Differentiators

Differentiator	Practical Contractor Pain Point Solved
Bilingual Voice-to-Form	Contractors on dusty sites with greasy hands can speak in Hindi or English to create sites without tedious typing.
Photo Register OCR Scanner	Allows taking a photograph of handwritten paper diaries, converting worker names and wages into digital records automatically.
Persistent Dues Tracker	Decouples physical construction completion from financial settlement; supplier credit doesn't disappear when work ends.
100% Offline Resilience	Full functionality in basements, remote road projects, and areas with zero mobile network reception.

Tool Overdue Warnings	Eliminates unnecessary tool rental costs by warning the contractor before equipment exceeds its return deadline.